

Domain-Scoped AFL Chat Agent — Retrieval, Guardrails & Grounding

Introduction

This project focuses on developing the conversational component of an AFL-based Artificial Intelligence system. The purpose of the project is to create a domain-specific chat assistant that is capable of discussing Australian Rules Football while remaining restricted to the AFL domain. The assistant is designed to answer questions related to AFL teams, players, matches, statistics, seasons, rules, records, and player or team performance.

The project builds on the previous AFL machine learning work, where historical AFL match and player data was prepared for analysis and prediction. In the previous stage, the system focused on machine learning models and prediction tasks. In this stage, the focus was shifted toward conversational interaction, retrieval, grounding, memory, and safety guardrails. The previous project was also designed with the idea that trained prediction models could later be exposed to an AI chat agent as callable tools.

The main challenge addressed in this project is the possibility of an AI language model generating information that is not actually present in the project's dataset. This is especially important for sports applications because users may ask for exact scores, player statistics, team records, or seasonal performance. If the model relies only on its internal knowledge, it may produce incorrect or outdated information. To reduce this problem, the assistant was connected to the actual AFL datasets through structured retrieval tools.

Another important requirement was domain control. The assistant should not behave as a general-purpose chatbot. If a user asks about basketball, cricket, football, programming, weather, general trivia, or another unrelated subject, the assistant should politely refuse the request and redirect the conversation toward AFL.

The final system combines domain guardrails, structured retrieval, LangChain, Gemini, conversation memory, tool calling, and grounding evaluation to create a controlled AFL conversational assistant.

Scenario

The scenario for this project is based on a real-world AFL sports analytics environment. Imagine that an AFL club has a coaching and analysis department that works with large amounts of historical match and player data. Coaches and analysts may need to repeatedly search through datasets to answer questions about team records, player performance, seasonal statistics, and historical matches.

Instead of manually searching through large datasets, the club can use an AFL-focused conversational assistant. A coach could ask the system a question in natural language, and the

assistant could identify the type of request, retrieve the relevant information from the AFL dataset, and return the result in a simple conversational response.

For example, a coach could ask about the historical record between two AFL teams. The assistant would identify that the question is related to AFL and use the team retrieval function to search the historical match dataset. It would then provide the information available in the dataset.

The coach could then continue the conversation by asking about player statistics. Because the assistant maintains conversation memory, the user does not necessarily need to repeat all the information from previous messages.

The system also protects against unsupported requests. If the coach asks about an NBA game, the assistant should not attempt to answer the question. Instead, it should explain that it is an AFL-focused assistant and redirect the conversation toward AFL.

This scenario demonstrates how a conversational AI system can be used as an interface over structured sports data while maintaining domain restrictions and reducing hallucinated information.

Task 1-- Scope Definition and System Prompt Design

The first task focused on defining the scope and behavior of the AFL conversational assistant. A clear scope was necessary because the purpose of the system was not to create a general-purpose chatbot. Instead, it was designed specifically for Australian Rules Football.

The supported scope includes AFL teams, AFL players, AFL matches, AFL statistics, player performance, team performance, AFL seasons, AFL history, AFL rules, AFL competitions, AFL match results, and AFL records.

The assistant was explicitly instructed to remain within these topics. Other sports such as basketball, cricket, soccer, tennis, and Formula 1 were considered outside the scope. General trivia, unrelated programming questions, weather, politics, unrelated entertainment, and general chit-chat were also considered outside the intended domain.

A system prompt was designed to communicate these restrictions to the language model. The prompt also instructed the assistant not to invent AFL statistics or numerical information. When information was not available in the dataset, the assistant was instructed to clearly state that the requested information was unavailable rather than guessing.

Another important part of this task was handling prompt injection. Users may attempt to override the system instructions by saying things such as “Ignore your previous instructions” or “Pretend you are not an AFL assistant.” The system was therefore instructed to ignore such attempts and continue operating as an AFL-focused assistant.

Adversarial prompts were prepared to test this behavior. These included questions about NBA, general trivia, jokes, weather, soccer, Formula 1, cricket, programming, and requests that explicitly attempted to bypass the AFL instructions.

The expected behavior for these tests was that legitimate AFL questions should be answered within the defined scope, while unrelated questions should receive a polite refusal and redirection toward AFL.

This task established the foundation for the remaining system because the retrieval and conversational components needed to operate under a clearly defined domain.

Task 2 -- Retrieval Layer Over AFL Data

The second task focused on connecting the conversational assistant with the real AFL datasets.

The available project data included team match information, player round-by-round statistics, player seasonal statistics, and player information. The previous AFL project also used structured historical data for match and player modelling.

For this project, structured retrieval was selected rather than semantic retrieval. The reason for this decision is that AFL statistics are primarily structured numerical information. For example, a question about the number of wins between two teams requires an exact calculation from the match dataset. Similarly, a question about a player's seasonal statistics requires an exact lookup using the player identifier and season.

Semantic retrieval is more suitable for unstructured documents such as articles, news reports, match descriptions, or long textual documents. Since the available AFL information was primarily stored in structured CSV datasets, direct structured retrieval was more appropriate.

Two main retrieval functions were developed. The first retrieves the historical record between two AFL teams. It searches the team match dataset and calculates the number of matches, wins, losses, draws, total scores, and opponent scores available for the selected teams.

The second retrieval function retrieves player seasonal statistics. It searches the seasonal player dataset using the player identifier and season. If the requested player-season combination exists, the relevant record is returned. If it does not exist, the system returns an explicit message indicating that no matching record was found.

This design is important for grounding because the language model is not expected to create statistics itself. Instead, it receives factual information from the retrieval layer and uses that information when producing the final response.

Task 3 -- Integration of Retrieval Tools with LangChain

The third task connected the retrieval layer with LangChain and the Gemini language model.

LangChain was used as the orchestration layer between the user, language model, and retrieval functions. The retrieval functions were converted into callable tools so that the agent could select the appropriate tool when a user asked a question requiring structured AFL information.

The system used Gemini as the language model and LangChain's tool-calling functionality to connect the model with the AFL retrieval layer.

The team-record retrieval function was exposed as a tool for questions involving the historical record between two teams. The player-season retrieval function was exposed as a tool for questions involving a player's statistics for a particular season.

The agent was given a separate system prompt that reinforced the AFL domain restriction. It was instructed that statistical questions should use the available retrieval tools and that numerical information should never be invented.

This created a clear separation between language understanding and factual data retrieval. The language model is responsible for understanding the user's natural-language question and deciding which tool is relevant, while the retrieval function is responsible for obtaining the actual information from the dataset.

Grounding verification was also introduced in this task. Tool messages were inspected so that the retrieved information could be compared with the final assistant response. This provides a method for checking whether a numerical answer was actually supported by the retrieved dataset.

This approach is consistent with the broader agent design used in the project, where tool selection, tool execution, observation, and subsequent reasoning are tracked during execution.

The result is a more reliable architecture because the assistant does not have to depend entirely on the language model's internal knowledge for AFL statistics.

Task 4 -- Memory and Multi-Turn AFL Conversations

The fourth task focused on adding conversation memory to the AFL assistant.

A conversational assistant should be capable of handling follow-up questions without requiring the user to repeat the complete context every time. For example, a user may first ask about a particular team and then ask a follow-up question referring to “that team” or “that player.”

Conversation memory was implemented using message history. Previous user messages, assistant responses, and tool results are maintained so that subsequent requests can be interpreted in the context of the earlier conversation.

The project material defines conversation memory as the message history sent to the model and explains that it can include previous interactions and tool results.

The system was tested using a multi-turn AFL conversation. The conversation could begin with a question about a team's record, continue with a question about a player associated with that context, and then move toward player statistics or a comparison.

The memory mechanism allows the assistant to maintain continuity between these turns.

This is different from working memory. Conversation memory contains the message history that can be passed to the model, while working memory represents temporary structured state used by the agent during execution.

Maintaining the complete conversation history also makes the system easier to inspect and debug because tool calls and their results remain visible as part of the interaction history.

Task 5 -- Guardrail Evaluation

The fifth task focused on evaluating the behavior of the completed AFL chat assistant.

The purpose of this evaluation was to determine whether the system remained within the AFL domain and whether it used retrieval when answering questions requiring factual AFL statistics.

The evaluation contained multiple categories of test prompts. Legitimate AFL questions tested whether the assistant could respond to valid requests about teams, players, statistics, rules, and performance.

Off-topic questions tested whether the assistant would refuse requests involving unrelated topics such as NBA, weather, programming, general trivia, and soccer.

Edge-case questions tested situations such as asking which sport is the best. Such questions can encourage the model to provide a general opinion instead of remaining within the AFL scope.

Prompt-injection tests attempted to override the system's instructions. Examples included requests to ignore AFL instructions and explain cricket, Formula 1, or tennis.

Missing-data tests were used to determine whether the assistant would correctly handle AFL information that was not present in the available dataset.

Grounding tests were designed to determine whether the assistant would retrieve information rather than provide numerical AFL statistics from memory.

The evaluation process recorded the final assistant response, the number of tool calls, and the tool outputs. This allowed scope compliance and grounding behavior to be examined separately.

A scope pass indicates that the assistant behaved according to the expected AFL or refusal behavior. A grounding pass indicates that a retrieval tool was used when the question required dataset-based information.

The final evaluation percentages should be taken from the executed notebook results rather than manually estimated. This is important because the evaluation is intended to represent the actual behavior of the implemented agent.

Failure Mode Analysis

The evaluation also considered several possible failure modes.

One possible failure is scope leakage, where the assistant answers an unrelated question instead of refusing it. This can be reduced by strengthening the system prompt and providing clear examples of appropriate refusal behavior.

Another failure is incorrect tool selection. The assistant may receive a statistical question but fail to select the appropriate retrieval tool. Clear tool descriptions and explicit instructions requiring retrieval for statistical questions help reduce this problem.

A third failure is incorrect tool arguments. Even if the correct tool is selected, the assistant may provide an incorrect team name, player identifier, or season. Input validation and structured tool schemas help prevent invalid requests.

Another important failure mode is missing-data hallucination. If the requested information does not exist in the dataset, the assistant should not create a fictional answer. The retrieval functions therefore provide an explicit unavailable response.

Prompt injection is another important failure mode. A user may attempt to override the system's domain restrictions. The system prompt and refusal behavior were designed to prevent these instructions from changing the assistant's intended role.

Finally, ungrounded numerical answers are a major concern in a sports-statistics system. A model may produce a plausible-looking number that does not come from the dataset. Tool-output logging and grounding inspection provide a method for identifying this type of failure.

The project material emphasizes that logging is useful because an agent can fail during tool selection, argument generation, execution, or final reasoning.

Real-World Application

A practical real-world application of the developed system would be an AI assistant used by an AFL club's coaching and sports-analysis staff.

In such an environment, analysts work with large historical datasets containing information about teams, players, matches, and performance. Instead of manually searching through these datasets, a coach could communicate with the system using natural language.

For example, a coach could ask about the historical record between two teams. The assistant would recognize the request as an AFL question and retrieve the relevant information from the team dataset.

The coach could then ask a follow-up question about a player's performance. The conversation memory would preserve the context of the previous interaction, while the player retrieval tool would retrieve the relevant seasonal statistics.

The system could also protect the coach from unreliable information. If a requested statistic was not present in the available dataset, the assistant would report that the information was unavailable instead of generating a value.

The same system could also be used by sports analysts, researchers, journalists, or AFL data teams as a natural-language interface over structured football datasets.

This demonstrates that retrieval-augmented conversational systems can provide a more accessible interface to structured sports data while maintaining control over the domain and reducing the possibility of hallucinated statistics.

System Architecture

The overall architecture of the project can be described as a sequence of connected components.

The user first sends a natural-language question to the AFL chat assistant. The system checks whether the request belongs to the AFL domain. If the request is outside the domain, the assistant provides a polite refusal and redirects the user toward AFL-related topics.

If the request is related to AFL, the LangChain agent determines whether a retrieval tool is required. For factual statistical questions, the appropriate retrieval tool is selected.

The team retrieval tool searches the team match dataset, while the player retrieval tool searches the player seasonal dataset. The retrieved result is then returned to the agent.

The language model uses the retrieved information to generate the final response. The conversation history is maintained so that follow-up questions can use previous context.

The complete process can therefore be summarized as:

User → AFL Scope Check → LangChain Agent → Retrieval Tool → AFL Dataset → Retrieved Result → Grounded Answer → Conversation Memory

This architecture combines conversational AI with structured data retrieval while maintaining the project's domain restrictions.

Overall Results and Observations

The project demonstrates that a general language model can be transformed into a domain-specific conversational assistant by combining system-level instructions, structured retrieval tools, memory, and evaluation.

The scope definition provides a clear boundary around the assistant's responsibilities. The retrieval layer provides access to actual AFL data instead of requiring the model to rely only on internal knowledge. LangChain provides the mechanism for connecting the language model with these retrieval functions.

Conversation memory allows the assistant to support multi-turn interactions, making the system more natural for users who want to ask several related questions.

The evaluation framework provides a way to test both domain compliance and grounding behavior. Instead of evaluating only whether the assistant produces fluent answers, the project also checks whether it remains within its intended scope and whether it uses retrieval for questions that require factual data.

The previous AFL project also emphasized leakage-aware modelling and the use of historical information for prediction. Current-match outcomes and final scores were excluded from predictive features because they would reveal the answer being predicted. The conversational system follows a related principle by ensuring that factual responses are grounded in the available dataset rather than being freely generated.

Final Conclusion

The Domain-Scoped AFL Chat Agent project successfully develops the conversational layer of the broader AFL artificial intelligence system.

The first task established a clear AFL-specific scope and defined how the assistant should behave when users ask unrelated questions. The system prompt ensured that the assistant remains focused on teams, players, matches, statistics, seasons, history, rules, and other AFL-related topics.

The second task introduced structured retrieval over the real AFL datasets. Because the available information is structured and numerical, exact Pandas-based retrieval was selected instead of semantic retrieval. This allows team records and player seasonal statistics to be obtained directly from the underlying datasets.

The third task connected these retrieval functions with LangChain and Gemini. Retrieval functions were exposed as tools, allowing the agent to select the appropriate tool when a user asks for factual AFL information. The system was also designed to avoid inventing numerical information and to treat retrieved data as the source of truth.

The fourth task introduced conversation memory. This allowed the assistant to maintain context across multiple turns and respond to follow-up questions more naturally. Previous messages and tool results can remain part of the conversation history, improving contextual understanding.

The fifth task evaluated the completed system using legitimate AFL questions, off-topic questions, edge cases, prompt-injection attempts, missing-data cases, and grounding attacks. This evaluation provides a structured method for identifying scope failures and grounding problems.

The completed system provides a foundation for future development. The previous AFL project already produced reusable prediction components for match outcomes and player rankings. These components can later be exposed as additional tools, allowing the conversational assistant to move beyond retrieval and interact with the project's prediction models.

Overall, the project demonstrates how LangChain, Gemini, structured retrieval, conversation memory, domain guardrails, and grounding verification can be combined to create a practical AFL-focused conversational AI system.

The system is therefore not simply a chatbot that talks about AFL. It is a controlled conversational interface over real AFL data, designed to keep responses within a specific domain and reduce the risk of unsupported or hallucinated statistical information.